PayBridge Architecture Decision Dossier

Keep remittance orchestration separate from ledger journals and use stable references to reconcile simulated payout attempts.

Executive Decision Summary

A repeat request should never add a second financial effect. The evidence to check is the remittance state, payout reference and journal balance for each currency.

The scope, boundaries, alternatives and evidence are documented in Parts A through E.

A Scope and requirements

Customers create remittance intents and convert with an FX quote. Treasury confirms simulated funding, triggers processing and recovers uncertain payout results. Monetary amounts use integer minor units and the workflow distinguishes customer access from treasury operations.

B Architecture and trust boundaries

Keycloak supplies JWTs to Spring Boot. The remittance and FX modules coordinate state and conversion. Ledger services persist journals and account movements. The local payout adapter simulates provider responses. The remittance controllers run in local and test profiles.

C Decisions and reversal signals

Use an idempotency key for create requests and a stable payout reference for recovery. An unknown result leaves the payout pending with float reserved. Treasury can reconcile it through operator-triggered recovery. Balanced journals separate funding, FX, reservation, release and payout by currency. A real provider would require reconciliation, finality and reversal policies before rollout.

D Verification and rollout

The implementation repository records workflow checks for replay, failed or unknown payout, recovery and journal balance. This review checked controllers, the response record and the documented state transitions. Application tests were not rerun for this design dossier. The contract focuses on the local remittance controllers and assumes an existing quote.

E AI use and evidence

Codex drafted the diagrams and dossier from public code. Customer and treasury paths, role requirements, profile gates and request records were checked against the controllers. Evidence about provider outcomes refers to the simulated adapter.